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**Department of Information Technology**

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Class: BE-IT A/B, Semester: VIII
Subject: Cloud Computing Lab

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# Experiment – 8: Security as a Service (SECaaS)

**Aim:** To study and Implement Security as a Service on AWS using ADDS service. (Free Tier Cloud Platforms).

**Objective:** After performing the experiment, the students will be able to –

- to understand Cloud security at AWS
- AWS Directory Service for identity management
- deploy AWS administration and security services.

**Lab objective mapped: ITDO8024.4**: To learn cloud deployment model and experiment security services over different cloud platforms.

**Prerequisite:**
An AWS Account

**Requirements:** Cloud Login**,** Desktop, Browser, Internet etc.

**Pre-Experiment Theory:**

**AWS Directory Service**
AWS Directory Service is a managed service that provides directories that contain information about your organization, including users, groups, computers, and other resources.

**Windows Server Manager**
Microsoft Windows Server Manager is a tool to view and manage server roles and make configuration changes. Server Manager allows administrators to manage local and remote servers without requiring physical access to the servers or enabling Remote Desktop Protocol connections.

**Procedure**
AWS Managed Microsoft AD prerequisites

- To create a AWS Managed Microsoft AD directory, you need a VPC with the following:
- At least two subnets. Each of the subnets must be in a different Availability Zone.
- The VPC must have default hardware tenancy.
- You cannot create a AWS Managed Microsoft AD in a VPC using addresses in the 198.18.0.0/15 address space.

To create an AWS Managed Microsoft AD directory

1. AWS Directory Service console navigation pane, choose Set up directory.
2. Select directory type : choose AWS Managed Microsoft AD, and then click Next.
3. Enter directory information page :
4. Edition :Choose Standard Edition.
5. Directory DNS name :The fully qualified name (FQDN) for the directory, *beit_exp5.com*.
6. Directory NetBIOS name : optional.
7. Directory description : An optional description for the directory.
8. Admin password : The password for the directory administrator. The directory creation process creates an administrator account with the user name **Admin**. (The password cannot include the word "admin."). Read rules for setting up the directory administrator password.
9. Confirm password
10. Choose VPC and subnets page (refer to VPC creation procedure) :
11. The VPC for the directory.
12. Subnets: Choose the subnets for the domain controllers. The two subnets must be in different Availability Zones.
13. choose Next.
14. On the Review & create page : review the directory information. Choose Create directory.
15. *Creating the directory takes 20 to 40 minutes (What gets created).*
16. *Once created, the Status value changes to Active (green in color).*

To seamlessly join a Windows EC2 instance

1. Go to Amazon EC2 console
2. From the Region selector in the navigation bar, choose the same Region as the existing directory.
3. Launch Instance.
4. AMI : win server 2019 free tier full edition
5. instance type : t2.micro
6. Next: Configure Instance Details.
7. Step 3 page:
8. Network : VPC that your directory was created in.
9. Subnet : choose one of the public subnets in your VPC. (The subnet that you choose must have all external traffic routed to an internet gateway.)
10. Domain join directory : choose your domain from the list.
11. IAM role:
    a. Choose Create role.

b. Under Select type of trusted entity, choose AWS service.
c. service role will use : choose EC2.
d. Select your use case : choose EC2
e. Next: Permissions.
f. In the list of policies : 1. AmazonSSMManagedInstanceCore 2.AmazonSSMDirectoryServiceAccess policies
g. Choose Next: Tags(Optional)
h. choose Next: Review.
i. Role name : EC2DomainJoin
j. Choose Create role.

12. Go back to the Step 3 page. For IAM role, choose the refresh icon next to IAM role.Choose created role and leave the rest of the settings with default values
13. choose Next: Add Storage.
14. On both the Step 4 and Step 5 pages, leave the default settings or make changes as needed, and then choose the Next buttons.
15. On the Step 6 page, select a security group for the instance that has been configured to allow remote access to the instance from your network, and then choose Review and Launch.
16. Step 7 page, choose Launch, select a key pair, and then choose Launch Instance..

To install the Active Directory administration tools on Windows Server 2012 through Windows Server 2019

1. Open Server Manager from the Start screen by choosing Server Manager.
2. In the Server Manager Dashboard, choose Add roles and features,In the Add Roles and Features Wizard choose Installation Type, select Role-based or feature-based installation, and choose Next.
3. Under Server Selection, make sure the local server is selected, and choose Features in the left navigation pane.
4. In the Features tree, open Remote Server Administration Tools, Role Administration Tools, select AD DS and AD LDS Tools, scroll down and select DNS Server Tools, and then choose Next.
5. Review the information and choose Install.
6. When the feature installation is finished, the Active Directory Domain Services and Active Directory Lightweight Directory Services Tools are available on the Start screen in the Administrative Tools folder.

To create a user

1. Open the Active Directory Users and Computers tool. There is a shortcut to this tool in the Administrative Tools folder.
2. On the Action menu, click New, and then click User to open the new user wizard.
3. On the first page of the wizard, enter the values for the following fields, and then click Next.
4. First name : student name

5. Last name : student last name
6. User logon name : first name
7. type a temporary password in Password and Confirm Password. Make sure the User must change password at next logon option is selected. None of the other options should be selected. Click Next.
8. On the third page of the wizard, verify that the new user information is correct and click Finish.
9. The new user will appear in the Users folder.

Create a new VPC

1. VPC console
2. choose Launch VPC Wizard.
3. Under Step 1: Select a VPC Configuration, on VPC with a Single Public Subnet, choose Select.
4. Enter the following information into the wizard and choose Create VPC.

IP CIDR block
10.100.0.0/16

VPC name
BEIT_EXP5_VPC

Public subnet
10.100.0.0/24

Availability Zone
us-east-1a

Subnet name
exp5_Subnet1

Enable DNS hostnames
Leave default selection

Hardware tenancy
Default

It takes several minutes for the VPC to be created.

**Post-Experiments Exercise**

1. Name three directory types that AWS supports? (to be written in hand)
2. List any five permissions an Admin account has to perform the administrative activities for your organizational unit (OU)? (to be written in hand)

**Extended Theory:**

What is Windows Server Manager?(soft copy)

Windows Server Manager is a built-in tool that comes with Windows Server operating system. It gives administrators a single dashboard to manage and monitor their server without switching between multiple tools.Using Server Manager you can install or remove roles and features like Web Server (IIS), DNS Server, DHCP Server, Active Directory and many more. You can also monitor the health of your server by checking CPU usage, memory, disk and network

performance.It also allows you to manage multiple remote servers from one place which saves a lot of time. You can view event logs, configure firewall settings, manage storage and users all from the same window.In simple words think of it like a control panel of a server where everything you need is available in one screen. Instead of opening different tools for different tasks Server Manager brings all of them together in one place.

## .Results/Calculations/Observations:

## Post Experimental Exercise-

How to reset password for already created user in AWS AD service?

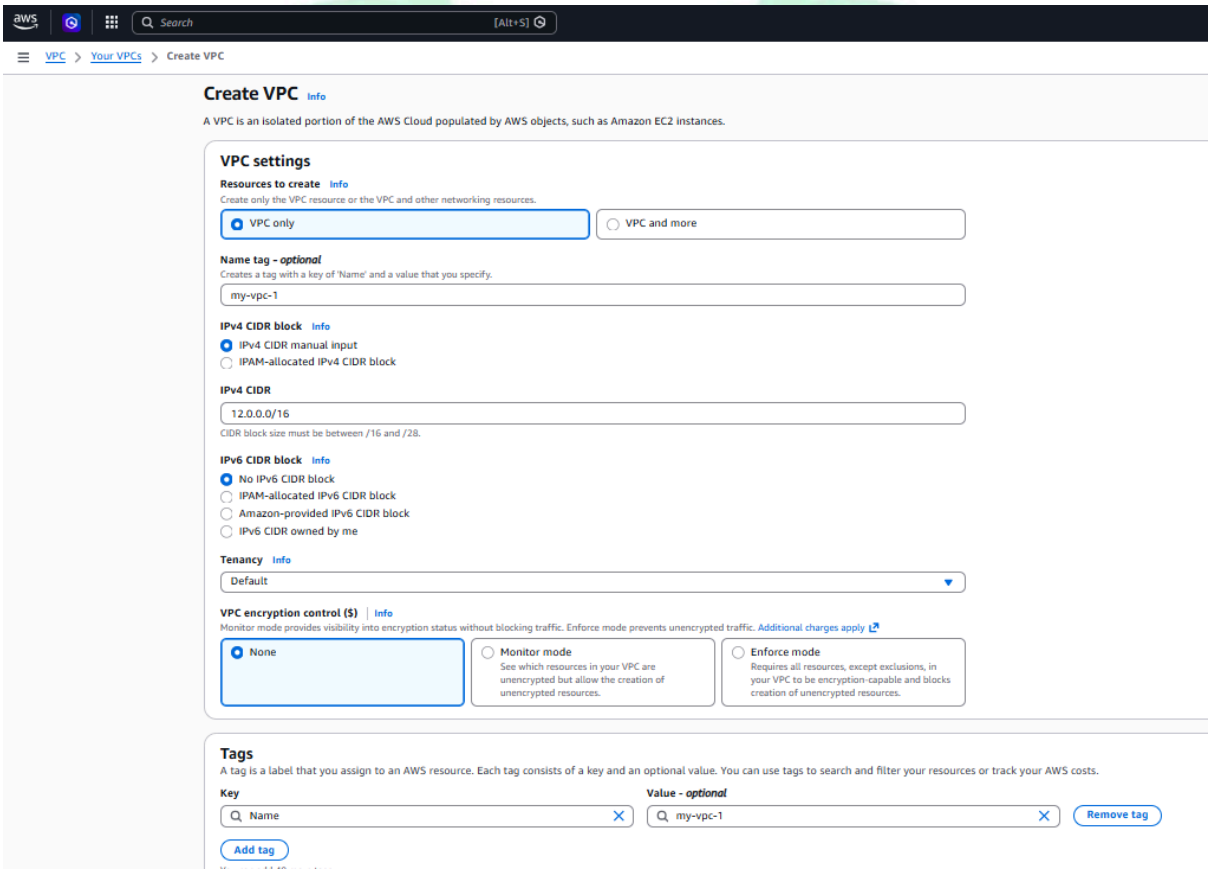
## Conclusion:

1. Write what was performed in the experiment
2. Mention a few applications of what was studied.
3. Write the significance of the studied topic

## References:


[1] [Online]https://learn.microsoft.com/en-us/azure/app-service

[2] [Online]https://learn.microsoft.com/en-us/azure/app-service/quickstart-nodejs

## Subnets (7) Info

Find subnets by attribute or tag

| Name | Subnet ID | State |
|---|---|---|
| soupnewt-2 | subnet-0aa38cdcd5dc4931a | Available |
| – | subnet-0460877a6a29474b0 | Available |
| – | subnet-0ad959ce345e0d1c8 | Available |
| aaronsubnet2 | subnet-05ade2ac40739894e | Available |
| – | subnet-07b5079813b88bddc | Available |
| aaronsubnet1 | subnet-07ad6ebc67bb4c413 | Available |
| soupnet-1 | subnet-08a21140c458b41f3 | Available |

VPC > Internet gateways > igw-02dd477e5caefba99

VPC dashboard

AWS Global View

Filter by VPC

Virtual private cloud

- Your VPCs
- Subnets
- Route tables
- Internet gateways
- Egress-only internet gateways
- DHCP option sets
- Elastic IPs
- Managed prefix lists

Internet gateway igw-02dd477e5caefba99 successfully attached to vpc-02bebbb9843e2d899

Notifications 0 0 2 0 0

## igw-02dd477e5caefba99 / gateaway

### Details Info

| Internet gateway ID | State | VPC ID |
|---|---|---|
| igw-02dd477e5caefba99 | Attached | vpc-02bebbb9843e2d899 \| vpc |

### Tags (1)

Search tags

| Key | Value |
|---|---|
| Name | gateaway |

VPC > Subnets > Create subnet

# Create subnet Info

## VPC

**VPC ID**
Create subnets in this VPC.

vpc-02bebbb9843e2d899 (vpc-1)

**Associated VPC CIDRs**

**IPv4 CIDRs**
12.0.0.0/16

## Subnet settings

Specify the CIDR blocks and Availability Zone for the subnet.

**Subnet 1 of 2**

**Subnet name**
Create a tag with a key of 'Name' and a value that you specify.

soupnet-1

The name can be up to 256 characters long.

**Availability Zone** Info
Choose the zone in which your subnet will reside, or let Amazon choose one for you.

No preference

**IPv4 VPC CIDR block** Info
Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must lie within this block.

12.0.0.0/16

**IPv4 subnet CIDR block**

12.0.1.0/24

VPC > Route tables > Create route table

# Create route table Info

A route table specifies how packets are forwarded between the subnets within your VPC, the internet, and your VPN connection.

## Route table settings

**Name - *optional***
Create a tag with a key of 'Name' and a value that you specify.

public-routetable

**VPC**
The VPC to use for this route table.

vpc-02bebbb9843e2d899 (vpc-1)

## Tags

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

| Key | Value - *optional* |
|---|---|
| Name | public-routetable |

Add new tag

You can add 49 more tags.

Search [Alt+S] Ask Amazon Q Asia Pacific (Mumbai) Swain

VPC > Route tables > rtb-0c1761ee99a985fad > Edit routes

# Edit routes

| Destination | Target | Status | Propagated | Route Origin | |
|---|---|---|---|---|---|
| 12.0.0.0/16 | local | Active | No | CreateRouteTable | |
| 0.0.0.0/0 | Internet Gateway<br>igw-02dd477e5caefba99 | – | No | CreateRoute | Remove |

Add route

Cancel Preview Save changes

EC2 > Instances > Launch an instance

**Key pair (login)** Info

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - *required*: pairkey — Create new key pair

**Network settings** Info

VPC - *required*: vpc-02bebbb9843e2d899 (vpc-1) 12.0.0.0/16

Subnet: subnet-08a21140c458b41f3 soupnet-1 — Create new subnet

Auto-assign public IP: Enable

Firewall (security groups): Create security group / Select existing security group

Security group name - *required*: launch-wizard-1

Description - *required*: launch-wizard-1 created 2026-04-06T05:06:36.278Z

Inbound Security Group Rules

Security group rule 1 (TCP, 22, 0.0.0.0/0) — Remove

| Type | Protocol | Port range | Source type | Source | Description - optional |
|---|---|---|---|---|---|
| ssh | TCP | 22 | Anywhere | 0.0.0.0/0 | e.g. SSH for admin desktop |

Security group rule 2 (All, All, 0.0.0.0/0) — Remove

| Type | Protocol | Port range | Source type | Source | Description - optional |
|---|---|---|---|---|---|
| All traffic | All | All | Anywhere | 0.0.0.0/0 | e.g. SSH for admin desktop |

**Summary**

Number of instances: 1

Software image (AMI): Amazon Linux 2023 AMI 2023.10....read more, ami-045445a70fa8b8bc

Virtual server type (instance type): t3.micro

Firewall (security group): New security group

Storage (volumes): 1 volume(s) - 8 GiB

Cancel — Launch instance — Preview code

ap-south-1.console.aws.amazon.com/ec2-instance-connect/ssh/home?addressFamily=ipv4&connType=standard&instanceId=i-0728ff590973

aws — Search [Alt+S]

```
  ,     #_
  ~\_  ####_        Amazon Linux 2023
 ~~  \_#####\
 ~~     \###|
 ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
  ~~       V~' '->
   ~~~         /
     ~~._.   _/
        _/ _/
      _/m/'
[ec2-user@ip-12-0-1-214 ~]$
```

ap-south-1.console.aws.amazon.com/ec2-instance-connect/ssh/hom

aws — Search [Alt+S]

**Failed to connect to your instance**

Error establishing SSH connection to your instance. Try again later.